

RES-5 endpoint projection factor: $\chi_{\text{proj}} = 1.25 > 1$ – the projection

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

res5-endpoint-2pi-bound v1.1 carried two closure levers for the $I = 2 \times 10^{-3}$ endpoint: the DR-2 floor κ , and the pattern projection $\chi_{\text{proj}} \leq 0.82$. This note tests the projection lever by directly computing the screened response at the pattern-modulation transfers. The result is a HONEST NEGATIVE: $\chi_{\text{proj}} = 1.25 > 1$, and the operator-norm tail estimate was an under-estimate. Scope: the $I = 2 \times 10^{-3}$ endpoint; the BCC $\{110\}$ modulation. No tier action.

2. The bubble is forward-peaked; screening is maximal at forward

The a0-skeleton response factor is $C_G = 1/(1 + g)$, $g = \lambda' \chi_0(0) = 1.03$, where $\chi_0(k) = \int \frac{d^3 p}{(2\pi)^3} G(p) G(|p + k|)$ is the bubble and $\chi_0(0) = \int q^2 G^2 / 2\pi^2 = B_d$. Direct evaluation gives $\chi_0(0) = 0.128$ and $\chi_0(k)$ MONOTONE DECREASING in $|k|$ (forward-peaked). Hence the screening factor

$$f(k) = \frac{1}{1 + \lambda' \chi_0(k)} \quad \text{is MINIMAL at } k = 0 \text{ (} f(0) = C_G = 0.49 \text{) and INCREASES with } |k|.$$

The screening C_G is the MAXIMAL screening (forward channel), not a generic bound.

3. The 110 modulation lives where screening is weaker

The modulation $\lambda'(P^2 - \langle P^2 \rangle)$ is supported on the BCC $\{110\}$ transfers $|t|/q_0 \in \{1, \sqrt{2}, \sqrt{3}, 2\}$, with weights w_t^2 (the same additive- energy structure as the floor). Every transfer is away from forward, so

$$f(t) \in \{0.567, 0.626, 0.679, 0.728\} > C_G = 0.492, \quad \frac{f(t)}{C_G} \in [1.15, 1.48].$$

The w_t^2 -weighted average is $f_{\text{avg}} = 0.613$, hence

$$\chi_{\text{proj}} = \frac{f_{\text{avg}}}{C_G} = \frac{0.613}{0.492} = 1.25 > 1 \quad (\text{not } < 0.82).$$

4. Verdict: projection route eliminated; tail corrected; solely DR-2-gated

Projection lever REFUTED. $\chi_{\text{proj}} = 1.25 > 0.82$: the res5-endpoint v1.1 projection lever is ELIMINATED. The modulation does not live in the maximally-screened channel, so the screening does not help beyond C_G – it helps LESS.

Tail CORRECTION. The operator-norm/screening estimate $C_{\text{higher}} \sim C_G a_0 = 0.047$ assumed forward screening; the corrected value is

$$C_{\text{higher}} = f_{\text{avg}} a_0 = 0.059 > 0.047.$$

This corrects the a0-skeleton's $C_G a_0$ upward by the factor $\chi_{\text{proj}} = 1.25$.

Endpoint now solely DR-2-gated. The corrected tail is still bracketed, $\text{slack}_{\text{proved}}(0.0385) < 0.059 < \text{slack}_{\text{verified}}(0.0758)$, so the endpoint closes ONLY at the verified floor, now with a 23% margin (was 38% under the under-estimate), and is OPEN at the conservative floor. The endpoint rests SOLELY on the DR-2 floor κ route; the second lever is gone. Off-endpoint ($I \leq 10^{-3}$) closure is UNAFFECTED – the $\geq 27\times$ margin absorbs the 1.25 factor (e.g. $I = 10^{-3}$: tail $1.25 \times 0.025 = 0.031 \ll \text{slack } 0.68$). No tier flip: B1-RH-ENUM stays T6 CONDITIONAL on {H-LAYER}.

5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “Is the bubble really forward-peaked? Brazovskii often has a $2q_0$ feature.” VALID-checked: the direct integral over $|k| \in [0, 3q_0]$ shows χ_0 monotone decreasing, no $2q_0$ peak, for the operating point $(\hat{r}, C, q_0)$; $g = \lambda' \chi_0(0) = 1.03$ reproduces the a0-skeleton value, confirming the normalisation. A sharper soft shell (larger C) could grow a $2q_0$ feature; at the operating point it is absent.
- (β) “Does this break the off-endpoint closure?” DISMISSED: no. The 1.25 factor multiplies the tail uniformly; for $I \leq 10^{-3}$ the slack is ≥ 0.68 and the tail ≤ 0.031 , ratio ≤ 0.05 – the off-endpoint strong- evidence closure (res5-tail-budget-closure) survives unchanged.
- (γ) “Then the endpoint is in worse shape than v1.1 claimed.” UPHELD: the endpoint margin thins from 38% to 23% and loses a lever; the honest gain is the elimination of a false closure route and the correct tail 0.059. The endpoint still closes at the verified floor; its sole rigorous residual is the DR-2 κ bound.

Quantitative sanity check (mandatory): *normalisation* $g = \lambda' \chi_0(0) = 1.03$ matches the a0-skeleton; *monotone* $\chi_0(q_0) = 0.095 < \chi_0(0) = 0.128$; *ratio* $f_{\text{avg}}/C_G = 0.613/0.492 = 1.25$; *corrected tail* $f_{\text{avg}} a_0 = 0.059 \in [0.0385, 0.0758]$ (verified-floor closure, 23%).

6. Result footer

Result ID: B1-RH-ENUM / RES-5 endpoint projection factor
Precise statement: $\chi_{\text{proj}} = f_{\text{avg}}/C_G = 0.613/0.492 = 1.25 > 1$
(not < 0.82).
The bubble $\chi_0(k)$ is forward-peaked, so the screening
 $f(k) = 1/(1 + \lambda \chi_0(k))$ is maximal at $k=0$
(= C_G) and weaker
at the {110} modulation transfers
($f=0.57-0.73$). The
operator-norm tail $C_G a_0=0.047$ is a
forward-channel under-
estimate; corrected tail $= f_{\text{avg}} a_0 = 0.059$.
The projection
closure lever ($\chi_{\text{proj}} \leq 0.82$) is ELIMINATED;
the endpoint
closes only at the verified floor
($0.059 < 0.0758$, 23%),
solely DR-2-gated. Off-endpoint closure
unaffected.

Scope: I=2e-3 endpoint; BCC {110} modulation;
operating anchor.

Dependencies: res5-endpoint-2pi-bound v1.1 (projection lever
tested here);
res5-a0-skeleton-sensitivity (C_G, a0,
corrected upward);
res5-tail-budget-closure (off-endpoint,
unaffected).

Evidence grade: ANALYTIC + EXECUTED (res5_projection_factor.py
5/5; direct
bubble integral + {110} additive-energy
weights).

Reproduction command: python codes/vacuum/res5_projection_factor.py

Expected output: chi0 forward-peaked; f(t)/C_G in [1.15,1.48];
chi_proj=1.25;
corrected tail 0.059 in [0.0385,0.0758]; 5/5
PASS.

Falsification gate: A bubble with a 2q0 peak above chi0(0) at the
operating
point (would lower chi_proj); or a {110} weight
shifted to
the forward channel.

Tier before / after: No tier flip. B1 T6 on {H-LAYER}. RES-5
endpoint: projection
lever ELIMINATED; tail corrected 0.047->0.059;
closure
solely via the DR-2 verified floor (23%
margin).

No-overclaim statement: This is a NEGATIVE for the projection route +
an upward tail
correction. The endpoint still closes at the
verified floor
but with a thinner margin and one fewer lever.
The a0-
skeleton C_G a0 estimate is corrected upward by
1.25. B1
unchanged.

Next required action: The endpoint's sole remaining lever is the
DR-2 floor kappa
bound (kappa<~0.8 worst-case over the dense
admissible
class). Pursue res5-dr2-kappa-bound.